REFINED DVR VALUATIVE CRITERION DETECTS PROPERNESS OVER A NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5121-5168, following the statement at 5104-5119.

Editorial note. No load-bearing omissions; the scheme-theoretic image replacement is displayed in the text.

Allowed standard prerequisites (editorial).

- Universally closed is etale-local on the base. Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-closed-local (cross-chapter, cited at ~5124-5126). Reduces to the affine etale base change X x_Y V -> V.
- Points of cartesian squares under surjective etale maps. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian (cross-chapter, cited at ~5127-5130). The image of |U x_Y V| is the preimage of the image of |h|.
- Etale morphisms are open. Properties of Spaces, Lemma spaces-properties-lemma-etale-open (cross-chapter, cited at ~5131-5133). Density of I in |X x_Y V|.
- Weak Chow's lemma for algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-weak-chow (cross-chapter, cited at ~5142-5144). Proper surjective X' -> X with X' a scheme.
- Valuative criterion characterizes proper morphisms of spaces. Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-proper (cross-chapter, cited at ~5157-5159). Lifts Spec(A) -> X to X' through the proper X' -> X.
- Scheme-level refined valuative criterion. Limits, Lemma limits-lemma-refined-valuative-criterion-proper (cross-chapter, cited at ~5162-5164). X' -> Y proper in the scheme case.
- Image of a proper morphism is proper. Morphisms of Spaces, Lemma spaces-morphisms-lemma-image-proper-is-proper (cross-chapter, cited at ~5165-5167). Concludes X -> Y proper.

Source reference. Original reference 0CMD: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 089J: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 6–6. Navigation pointer

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original reference 0A40: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

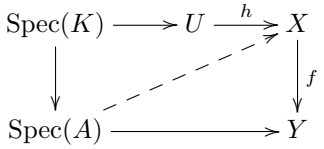
Source reference. Original reference 08AJ: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IT: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IR: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-limits-refined-valuative-criterion-proper/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $h : U \to X$ be morphisms of algebraic spaces over S. Assume that Y is locally Noetherian, that f and h are of finite type, that f is separated, and that the image of $|h| : |U| \to |X|$ is dense in $|X|$. If given any commutative solid diagram*



where A is a discrete valuation ring with field of fractions K, there exists a dotted arrow making the diagram commute, then f is proper.

Proof. It suffices to prove that f is universally closed. Let $V \to Y$ be an étale morphism where V is an affine scheme. By Morphisms of Spaces, Lemma 03IT it suffices to prove that the base change $X \times_Y V \to V$ is universally closed. By Properties of Spaces, Lemma 03H4 the image I of $|U \times_Y V| \to |X \times_Y V|$ is the inverse image of the image of $|h|$. Since $|X \times_Y V| \to |X|$ is open (Properties of Spaces, Lemma 03IR) we conclude that I is dense in $|X \times_Y V|$. Therefore the

assumptions of the lemma are satisfied for the morphisms $U \times_Y V \to X \times_Y V \to V$. Hence we may assume Y is an affine scheme.

Assume Y is an affine scheme. Then U is quasi-compact. Choose an affine scheme and a surjective étale morphism $W \to U$. Then we may and do replace U by W and assume that U is affine. By the weak version of Chow's lemma (Cohomology of Spaces, Lemma 089J) we can choose a surjective proper morphism $X' \to X$ where X' is a scheme. Then $U' = X' \times_X U$ is a scheme and $U' \to X'$ is of finite type. We may replace X' by the scheme theoretic image of $h' : U' \to X'$ and hence $h'(U')$ is dense in X'. We claim that for every diagram

$$\begin{array}{ccccc} \mathrm{Spec}(K) & \longrightarrow & U' & \xrightarrow{h} & X' \\ \downarrow & & & \nearrow & \downarrow{\scriptstyle f'} \\ \mathrm{Spec}(A) & & \longrightarrow & & Y \end{array}$$

where A is a discrete valuation ring with field of fractions K, there exists a dotted arrow making the diagram commute. Namely, we first get an arrow $\mathrm{Spec}(A) \to X$ by the assumption of the lemma and then we lift this to an arrow $\mathrm{Spec}(A) \to X'$ using the valuative criterion for properness (Morphisms of Spaces, Lemma 0A40). The morphism $X' \to Y$ is separated as a composition of a proper and a separated morphism. Thus by the case of schemes the morphism $X' \to Y$ is proper (Limits, Lemma 0CMD). By Morphisms of Spaces, Lemma 08AJ we conclude that $X \to Y$ is proper. □